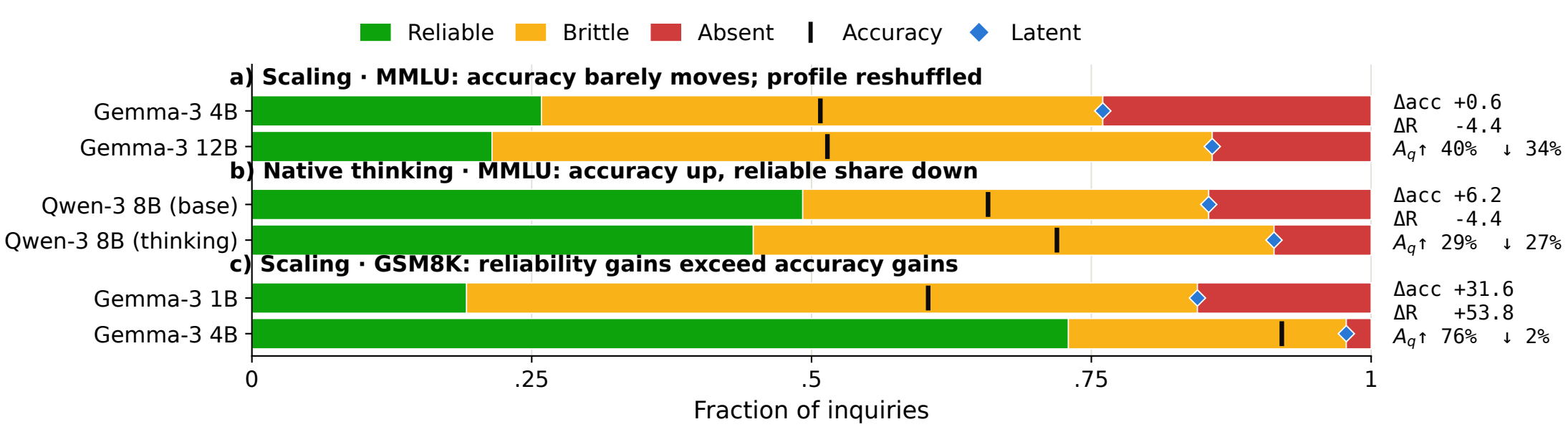
Reliable
Brittle
Absent
Accuracy
Latent
a) Scaling · MMLU: accuracy barely moves; profile reshuffled
Gemma-3 4B
Gemma-3 12B
b) Native thinking · MMLU: accuracy up, reliable share down
Qwen-3 8B (base)
Qwen-3 8B (thinking)
c) Scaling · GSM8K: reliability gains exceed accuracy gains
Gemma-3 1B
Gemma-3 4B
Δacc +0.6
ΔR -4.4
A_q↑ 40% ↓ 34%
Δacc +6.2
ΔR -4.4
A_q↑ 29% ↓ 27%
Δacc +31.6
ΔR +53.8
A_q↑ 76% ↓ 2%
0
.25
.5
.75
1
Fraction of inquiries